HI, I'M

Rafael Antunes Vieira

Head of Technology · Software Architecture · Technical Leadership

I lead agile teams and design scalable architectures for mission-critical systems. Five years building and integrating **SICAR Federal**, Brazil's National Rural Environmental Registry, with strong expertise in geoprocessing, PostgreSQL/PostGIS and cloud infrastructure.



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12+	5	QA → Head	GIS
years in technology	years on SICAR Federal	full journey across the software lifecycle	spatial data specialist

01 About me

I am a Head of Technology and Software Engineer with a solid track record managing agile teams and defining scalable architectures. I started in 2014 as an infrastructure technician, then moved through QA, full-stack development and technical leadership up to strategic technology management. This gives me an end-to-end view, from server to product.

I work on implementing Artificial Intelligence in enterprise systems and modernizing legacy applications. I have broad experience integrating complex ecosystems, including government projects such as **SICAR Federal** and the implementation of Brazil's **Forest Code**, plus strong skills in geoprocessing and cloud infrastructure.

My focus is delivering solutions that combine technological innovation, security and high performance, while staying hands-on in the code.

Leadership

Managing multidisciplinary teams, technical mentoring and strategic IT planning.

Architecture

Microservices, systems integration and legacy modernization with a focus on scale.

Spatial data

PostgreSQL/PostGIS, geoprocessing and high-volume environmental data.

02 Professional experience

Sep 2025 - Present

Entrega Contínua

Belo Horizonte, Brazil

Head of Technology

- Technical and strategic leadership of the technology area, defining system architecture and selecting the most suitable technologies for each project.
- Full management of server infrastructure and cloud environments, ensuring high availability (SLA), robust security and service scalability.
- Leading the development team, fostering technical mentoring, collaboration and professional growth.
- Hands-on participation in software development, from implementing critical features to running tests.
- Defining and tracking agile processes, performance metrics and engineering best practices.
- Driving technological innovation by identifying continuous-improvement opportunities in legacy systems and new products.

Dec 2024 - Jan 2026

IICA

Inter-American Institute for Cooperation on Agriculture · Remote

Senior IT Consultant

- Specialized technical consulting on integrating products with the National Rural Environmental Registry System (SICAR Federal).
- In-depth integration diagnostics, identifying interoperability failures and performance bottlenecks.
- Proposing and implementing architectural improvements and fixes in APIs and databases to ensure data compliance.
- Updating and writing technical integration manuals between state environmental management systems and SICAR Federal.

Aug 2022 - Sep 2025

Youx Group

Lavras, Brazil

Software Tech Lead Jan 2024 - Sep 2025

- Leading a multidisciplinary team dedicated to SICAR Federal, providing technical guidance and support on complex engineering problems.
- End-to-end management of the development lifecycle (SDLC) for SICAR's critical deliveries, from conception to production.
- Direct collaboration with internal and government stakeholders to align technical requirements and business rules.
- Administering PostgreSQL/PostGIS databases for the project's spatial data, with performance tuning and backup strategies.

Mid-level Systems Analyst Aug 2022 - Dec 2023

- Full-stack development building and maintaining SICAR Federal with JavaScript (AngularJS, Vue.js) and Java (Play Framework, Spring).
- Production troubleshooting and continuous evolution of the system's data structure.
- Planning and delivering technical training for users and advanced support to other project teams.

- Building and maintaining automated deployment pipelines with GitLab Runner for safe, continuous delivery.

Feb 2021 - Aug 2022

Agência Zetta

Lavras, Brazil

Junior Systems Analyst Feb 2022 - Aug 2022

- Developing routines and features for SICAR Federal with JavaScript (AngularJS, Vue.js) and Java (Play Framework, Spring).
- Identifying and fixing defects to guarantee environmental data integrity and continuous system operation.
- Writing technical and critical business-rule documentation to ease technology transfer.

Software Quality Analyst (QA) Feb 2021 - Feb 2022

- Running regression, functional, integration, usability and performance tests to ensure SICAR Federal's stability.
- Producing user manuals, use cases and detailed test scenarios.
- Tracking the lifecycle of bugs and improvements in GitLab and Mantis.

Feb 2014 - Dec 2020

Freelance

Piranguçu, Brazil

Infrastructure & Maintenance Technician

- Technical support for corporate clients, educational institutions and retail, as well as home visits.
- Installing, configuring and maintaining computer networks and internet access.
- Advanced maintenance and configuration of Linux and Microsoft Windows servers.

03 Featured projects

FEDERAL GOVERNMENT · 2021 - 2025

SICAR Federal: National Rural Environmental Registry System

The national platform that registers and analyzes Brazil's rural properties to enforce the Forest Code. I worked across the entire cycle, from QA to Tech Lead, building critical features, administering the PostgreSQL/PostGIS databases holding spatial data for the whole country, creating CI/CD pipelines and leading the multidisciplinary team responsible for deliveries.

Java | Spring | Play Framework | AngularJS | Vue.js | PostGIS | GitLab CI | Linux

CONSULTING · IICA · 2024 - 2026

Integrating state systems with SICAR

Diagnosing and fixing interoperability failures between the states' environmental management systems and SICAR Federal. Architectural improvements in APIs and databases and authoring of the technical integration manuals.

APIs REST · Interoperability · PostgreSQL · Technical documentation

LEADERSHIP · ENTREGA CONTÍNUA · 2025 - PRESENT

Legacy modernization and AI in enterprise systems

Defining the architecture and stack for new products, managing cloud infrastructure with a focus on SLA and security, and implementing Artificial Intelligence in existing enterprise systems.

Architecture · Cloud · Docker · Applied AI · Microservices

04 Skills

Leadership & Strategy

Multidisciplinary team management · Technical mentoring · Strategic IT planning · Scrum · Kanban · Stakeholder management

Software Engineering & AI

Java · Spring Boot · Play Framework · JavaScript · Angular · Vue.js · C++ · Microservices · AI in enterprise systems · Android

Infrastructure & Data

PostgreSQL · PostGIS · QGIS · Spatial data · Cloud Computing · Docker · CI/CD · GitLab Runner · Linux Server · Windows Server · Networking

Business Domain

Government systems · SICAR · Brazilian Forest Code · Systems interoperability · Complex business rules · Software quality

AWS Cloud

Amazon EC2 · Amazon S3 · AWS Lambda · Amazon RDS · AWS Trusted Advisor · AWS Health · AWS CloudTrail · Amazon CloudWatch · Amazon EFS · AWS Storage Gateway · AWS Elastic Disaster Recovery · Amazon EBS · Amazon SQS

05 Education & certifications

Academic background

2018 - 2022

B.Sc. in Computer Science

Federal University of Itajubá (UNIFEI) · Itajubá, Brazil

2014 - 2015

English as a Second Language (ESL)

Wizard by Pearson · Immediate Conversation, 140 hours

Certifications & courses

2026

AWS Cloud Practitioner Essentials

AWS Training & Certification · AWS cloud fundamentals: core services, security, architecture and pricing.

2026

Mastering Scrum

Entrega Contínua · 14-hour training.

2024

PostgreSQL/PostGIS Database Management with QGIS

Udemy · Administration, spatial data modeling and integration with geoprocessing tools.

2021

Complete Scrum Training: Agile Planning and Project Management

Unieducar · Traditional and agile management, Scrum principles, roles and responsibilities. 4 hours.

2021

Android Development: Learn to Build 15 Apps

Udemy · Native apps, activity lifecycle, REST API integration and data persistence. 54.5 hours.

06 Let's talk?

Open to technical leadership, software architecture and consulting opportunities in government systems and geoprocessing. I will reply as soon as possible.

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